(Advanced Drone Technology.)

INDIA SPACE LAB

Project Report

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TASK 1 – Understanding PID Controller Design Using Drone Altitude Control

Objective

The objective of this task was to understand how a PID (Proportional–Integral–Derivative) controller works by using a drone altitude control simulation. The experiment also examined how the controller maintained the desired altitude when wind disturbance was introduced.

Introduction

A PID controller is one of the most widely used feedback control algorithms in robotics and autonomous systems. It continuously compares the desired altitude with the actual altitude and adjusts the control input to reduce the error.

where:

- **Kp** = Proportional gain
- **Ki** = Integral gain
- **Kd** = Derivative gain
- **e(t)** = Desired altitude − Actual altitude

Software Used

- Google Colab
- Python
- NumPy
- Matplotlib
- ipywidgets

Procedure

1. Executed the provided notebook.
2. Tuned the PID controller to obtain a stable altitude response.
3. Introduced wind disturbance after 6 seconds.
4. Observed the drone's behaviour before and after the disturbance.
5. Selected the controller gains that provided the best overall performance.

Final Controller Parameters

Parameter	Value
Proportional Gain (Kp)	**2.30**
Integral Gain (Ki)	**0.91**
Derivative Gain (Kd)	**1.50**
Wind Noise	**2.00**

Observation

- The drone reached the target altitude of **10 m** quickly.
- A small overshoot was observed during the initial response.
- Oscillations gradually reduced as the controller stabilized.
- After the wind disturbance was introduced at **6 seconds**, the altitude changed only slightly.
- The controller corrected the error quickly and maintained a stable altitude close to the desired setpoint.

Results

The final controller achieved:

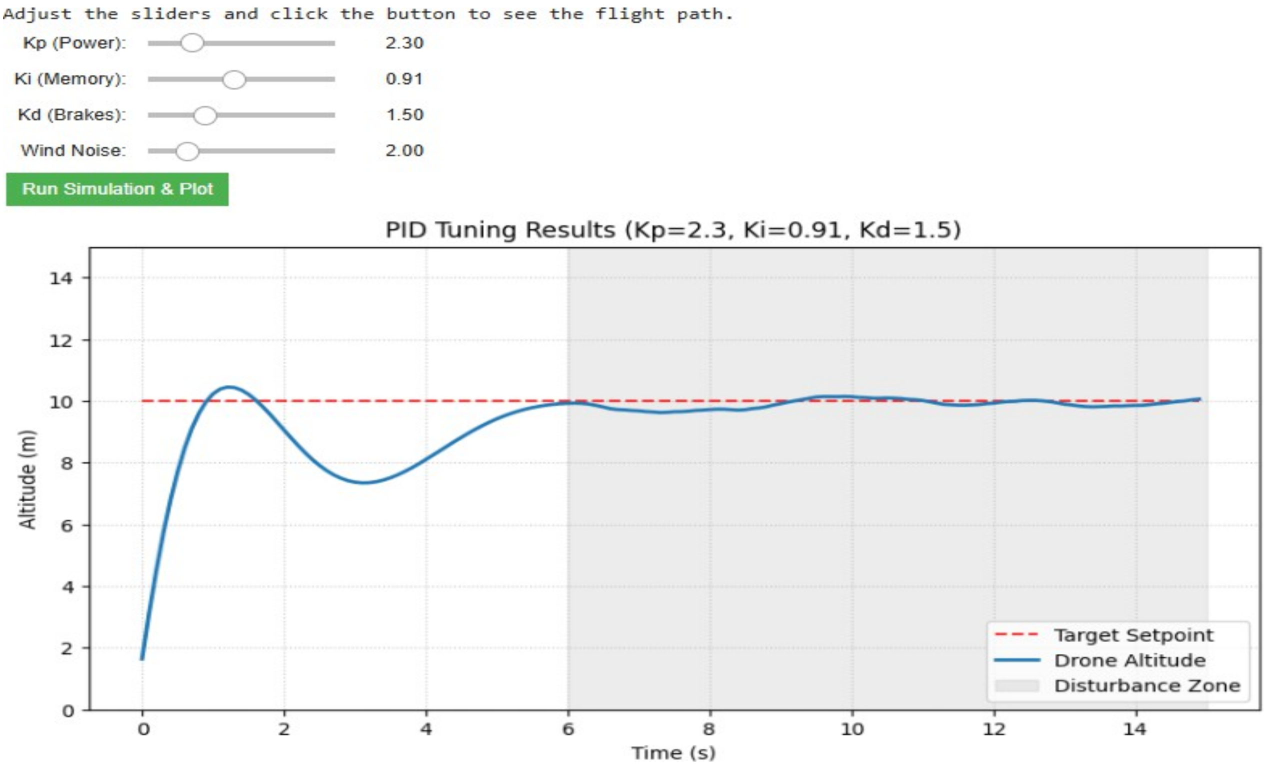
- Fast response to the target altitude.
- Small overshoot.
- Low oscillations.
- Stable altitude tracking.
- Good rejection of wind disturbance.

Conclusion

The PID controller successfully maintained the drone at the required altitude throughout the simulation. The selected gains (**Kp = 2.30, Ki = 0.91, Kd = 1.50**) produced a fast and stable response with minimal overshoot and effectively compensated for the wind disturbance.

Figure

Figure 1. PID Controller Tuning Result Plot (Kp = 2.30, Ki = 0.91, Kd = 1.50, Wind Noise = 2.00)



TASK 2 – Boat Guidance Control

Objective

The objective of this task was to evaluate the performance of a boat guidance controller in following a predefined path under both ideal conditions and in the presence of water current disturbances.

Introduction

The boat guidance controller is designed to minimise the tracking error while following a desired path. Its performance depends on the selected controller gain and damping values, as well as its ability to compensate for external disturbances such as water currents.

Software Used

- Google Colab
- Python
- NumPy
- Matplotlib
- ipywidgets

Procedure

1. Ran the boat guidance simulation.
2. Set the controller gain and damping values.
3. Observed the boat trajectory without water current.
4. Introduced water current disturbance.
5. Compared the tracking performance under both conditions.

Final Controller Parameters

Parameter	Value
Gain	**1.00**
Damping	**1.00**
Current X (Without Disturbance)	**0.00**
Current Y (Without Disturbance)	**0.00**
Current X (With Disturbance)	**0.12**
Current Y (With Disturbance)	**0.12**

Observation

Without Disturbance

- The boat followed the reference path smoothly.
- Tracking error was very small.
- Motion remained stable from the start to the end of the trajectory.

With Disturbance

- The applied water current caused a slight deviation from the desired path.
- The controller continuously corrected the boat's direction.

- The boat remained stable and continued to follow the reference trajectory despite the disturbance.

Results

- Smooth trajectory tracking under normal conditions.
- Small tracking error.
- Stable controller response.
- Good disturbance rejection when water current was applied.

Conclusion

Using **Gain = 1.00** and **Damping = 1.00**, the boat guidance controller achieved smooth trajectory tracking under normal conditions and remained stable even when water current disturbances were introduced.

Figures

Figure 2. Boat Guidance Plot without Disturbance (Current X = 0.00, Current Y = 0.00)

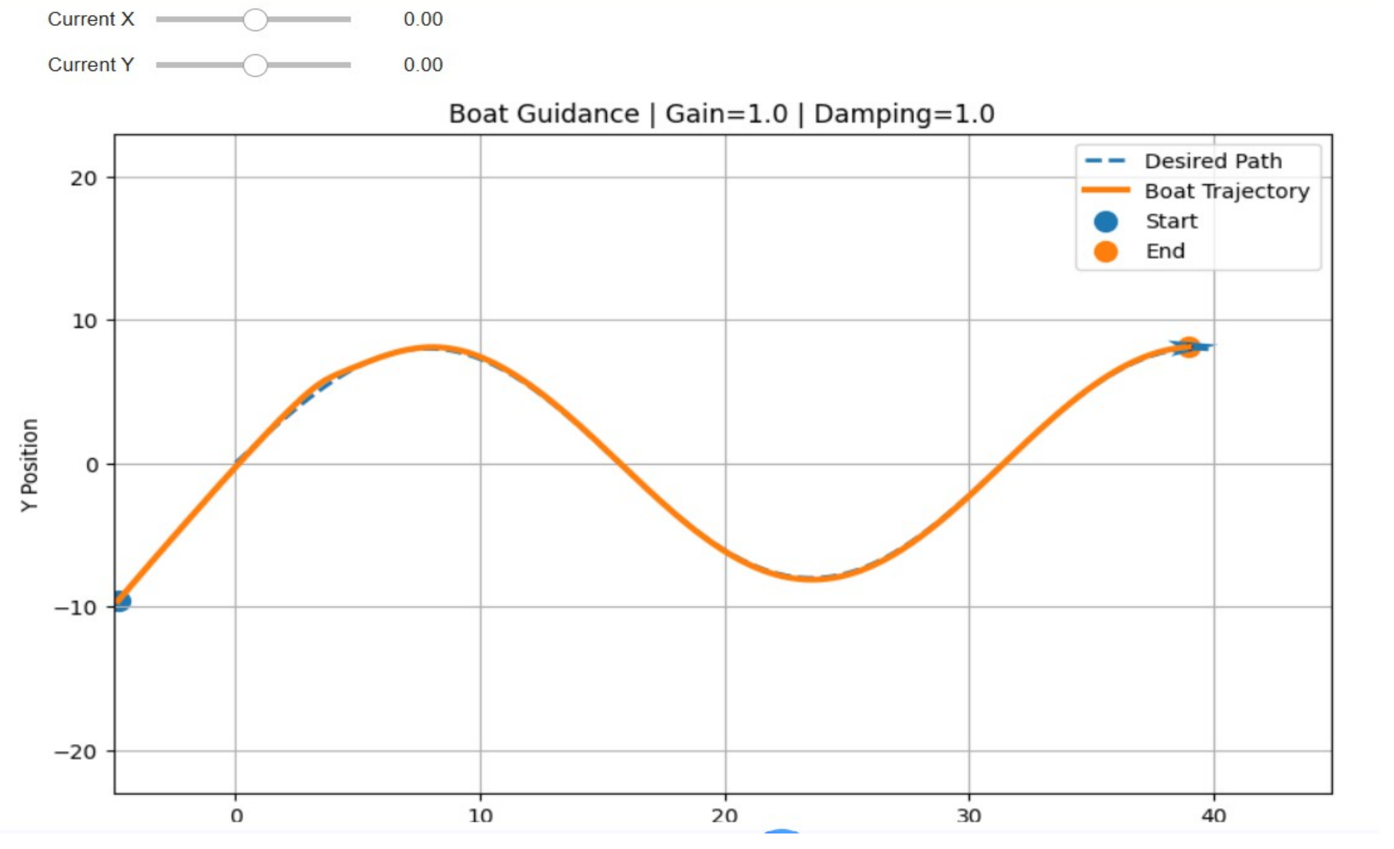
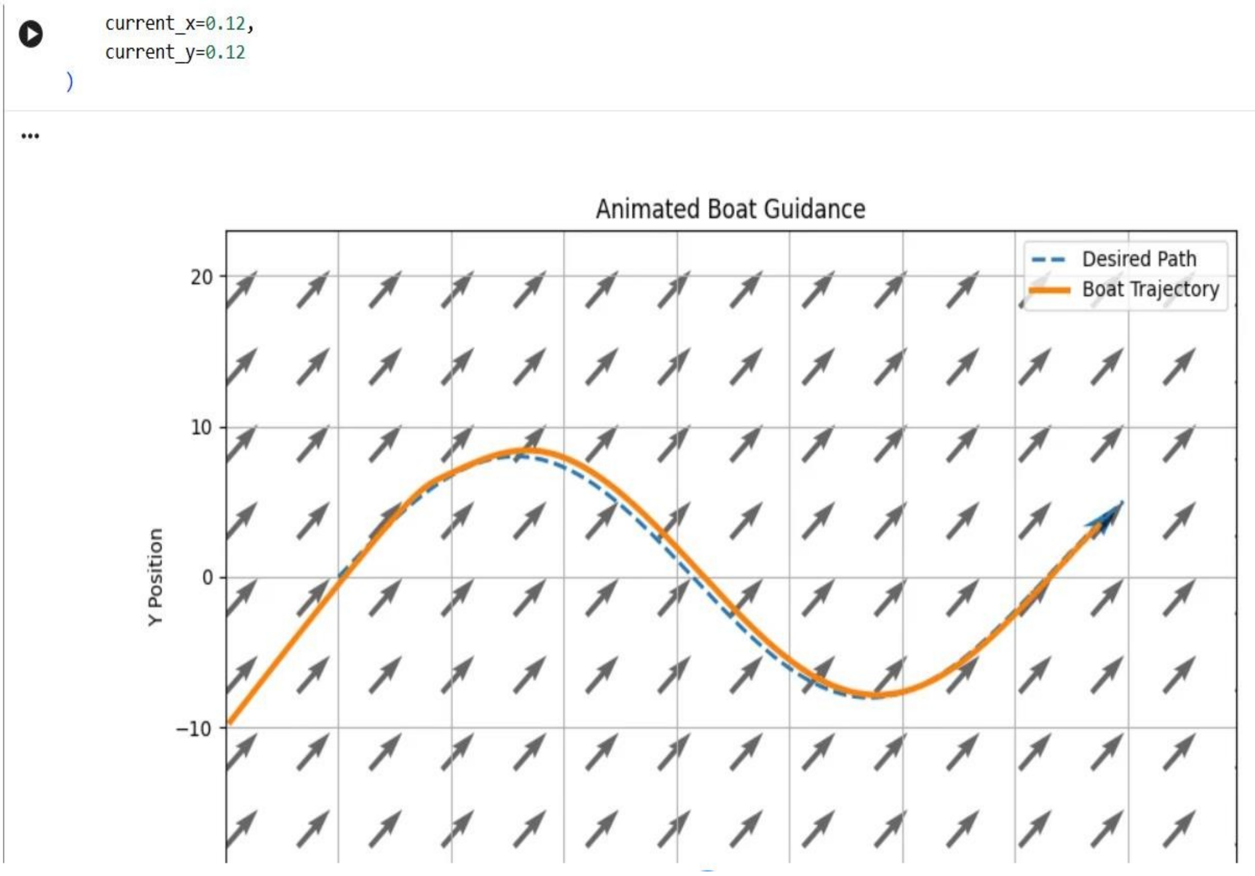


Figure 3. Boat Guidance Plot with Disturbance (Current X = 0.12, Current Y = 0.12)



TASK 3 – Figure-Eight Drone Simulation (Bonus Task)

Objective

The objective of this task was to develop a Python simulation in which a drone follows a figure-eight trajectory while maintaining smooth and stable flight.

Introduction

A figure-eight trajectory is commonly used to evaluate the path-following ability of autonomous drones because it requires continuous changes in direction. In this simulation, the trajectory was generated using parametric equations, and a tracking controller was used to minimise the tracking error.

The reference trajectory is defined as

x=Asin(t)x=A\sin(t)x=Asin(t) y=Bsin(t)cos(t)y=B\sin(t)\cos(t)y=Bsin(t)cos(t)

where

- **A** = Horizontal amplitude
- **B** = Vertical amplitude
- **t** = Time

Software Used

- Google Colab
- Python
- NumPy
- Matplotlib

Procedure

1. Generated a figure-eight reference trajectory.
2. Applied a tracking controller to follow the path.
3. Tested the drone under normal and disturbed conditions.
4. Observed the tracking performance and flight stability.

Observation

- The drone successfully followed the figure-eight path.
- The trajectory was smooth with gradual turns.
- Tracking error remained low throughout the simulation.
- The controller maintained stable flight even when disturbances were introduced.
- The drone quickly returned to the reference path after small deviations.

Results

The simulation demonstrated:

- Smooth figure-eight trajectory tracking.
- Stable flight throughout the simulation.
- Low tracking error.
- Good recovery from disturbances.

Conclusion

The implemented controller successfully guided the drone along the figure-eight trajectory while maintaining smooth and stable motion. The simulation demonstrated accurate trajectory tracking, low tracking error, and good overall control performance.

Figure

Figure 4. Figure-Eight Drone Trajectory Tracking Simulation